

Education

- 2023-27 **B.A.** Computer Science, **B.A.** Data Science*, **Minor** Bioengineering, *University of California, Berkeley*
- 2020-23 **IB Diploma** Higher level Math AA, Biology, Chemistry, *International English Gymnasium Sodermalm*

Research Experience

- 2025– ML Researcher, [Yun S. Song Lab](#), [Innovative Genomics Institute](#) & Berkeley Artificial Intelligence Research. (Advisors: Yun S. Song, Rohan Sachdeva, Spencer Diamond)
- 2023– Computational Immunology Researcher, Borch Lab, Washington University in St. Louis. (Advisor: [Nicholas Borcharding](#))
- 2024 Bioinformatics & AI engineer, [Generation Lab, Inc.](#) (YC S24)
- 2023-25 Researcher & Bioinformatician, Irina M. Conboy Lab, University of California, Berkeley & [QB3](#).
- 2023 Researcher, Hallucinating Scaffolds team, [iGEM at Berkeley](#). (Advisor: [Chris Anderson](#))
- 2023 NGS Method developer, [Sahlin Group](#), SciLifeLab & Stockholm University. (Advisor: [Kristoffer Sahlin](#))
- 2022-23 Researcher, [Karlsson-Hedestam/Murrell Lab](#), Karolinska Institute. (Advisor: [Benjamin Murrell](#))

Awards & Honors

- 2025 Dean's List - UC Berkeley College of Statistics, Data Science, and Society
- 2023 Finalist, National Swedish Research fair ([1/55](#))
- 2023 Top 60, National Swedish Programming Olympiad Site
- 2022 Top 115, National Swedish Chemistry Olympiad
- 2022 Double Admit to Karolinska Institute's summer research programs (1/5 admitted nationwide)
- 2022 Top school participant, Swedish Biology Olympiad

*With [Domain Emphasis](#) of Computational Methods in Molecular and Genomic Biology

Fellowships & Funding

- 2025 Top 20 Global Fellow, competitive BioTech/AI entrepreneurship program; eligible for mid six-figure funding (< 0.1% acceptance rate)
- 2025 Recipient, Anna Whitlock Scholarship, [Anna Whitlock Memorial Foundation \[In Swedish\]](#)
- 2023 Recipient, PostHS scholarship, [Equitable College Counselling](#)

Academic Service

- 2026 Program Committee Member, Chen Institute Symposium for AI Advancing Science & Society 2026
- 2026 Abstract Reviewer, Gerontological Society of America (GSA) Annual Scientific Meeting

Publications

† → Equal contribution

Journal Articles

- J1. **Yang, Qile**, Safina, K. R., Nguyen, K. D. Q., Tuong, Z. K. & Borcharding, N. scRepertoire 2: Enhanced and efficient toolkit for single-cell immune profiling. *PLOS Computational Biology* **21**, 1–10. <https://doi.org/10.1371/journal.pcbi.1012760> (June 2025).
– Media: [ASCA](#), [Chinese Software Developer Network \[In Chinese\]](#), [Jianshu \[In Chinese\]](#)
– BioRxiv preprint: <https://doi.org/10.1101/2024.12.31.630854>.
- J2. **Yang Qile**. APackOfTheClones: Visualization of clonal expansion with circle packing. *Journal of Open Source Software* **9**, 6868. <https://doi.org/10.21105/joss.06868> (Nov. 2024).
– Media: [R views \(by Rstudio\)](#), [R-bloggers](#), [Zhihu \[In Chinese\]](#), [360doc \[In Chinese\]](#), [Tencent \[In Chinese\]](#).
- J3. Cruz, J. M., Yeung, H., Alzalzalee, R., **Yang, Qile**, Kabir, H., McDonough, S., Mei, X., Conboy, M. J. & Conboy, I. M. In old mice, Exercise Induces Inflammation and Fibrosis unless Alk5-inhibitor and Oxytocin are used. *Journal of Cellular Physiology* **240**, e70054. <https://onlinelibrary.wiley.com/doi/abs/10.1002/jcp.70054> (June 2025).
- J4. Cruz, J. M. C., Alzalzalee, R., Yeung, H., Mahmood, Z., **Yang, Qile**, Morshedian, N., Robinson, Z. R., Malhotra, K. R., Conboy, M. J., Mazahery, A. R., Nevado, J. B. & Conboy, I. M. Plasma Dilution After Myocardial Ischemia–Reperfusion Injury Promotes Cardiac Repair, Heart Performance, and Recovery of Motor Function and Endurance in Old Mice. *Aging Cell* **25**, e70525 5709210, e70525. eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1111/ace1.70525>. <https://onlinelibrary.wiley.com/doi/abs/10.1111/ace1.70525> (Apr. 2026).

Working Papers

- W1. Borcharding, N., **Yang, Qile**, Sun, B., DeNardo, D., Mudd, P. A., Ellebedy, A. & Brestoff, J. R. *Ibex: Deep autoencoders for single-cell BCR sequencing* Manuscript under preparation. 2025. <https://github.com/BorchLab/Ibex>.

Presentations

Posters

- P1. **Yang, Qile**, Sahlin, K. & Murrell, B. *KmerGMA: A seed-based approach for fast homology searching* Swedish National Science Fair Finals (Södertälje, Sweden). Apr. 2023. <https://qile0317.github.io/KmerGMA.jl/stable/>.
- P2. **Yang, Qile** & Borcharding, N. *Reimagining Single Cell BCR Modeling: An Interpretable, Biochemically-Informed Framework* Cal Undergraduate Bioengineering Symposium. Dec. 2025. <https://igem.studentorg.berkeley.edu/cubs/>.
- P3. **Yang, Qile** & Borcharding, N. *Why use a pLM when Tiny Autoencoders will do? Dissecting Antibody Representations and Scaling to Single-Cell BCRs with Ibex* Bay Area Bioengineering Symposium. Apr. 2026. <https://igem.studentorg.berkeley.edu>.

Demonstrations & Tutorials

- D1. **Yang, Qile** & Murrell, B. *Discovery of Novel Camelid Germline Immunoglobulin Alleles* Karolinska Institute's Summer Research Program in Computational Biology (Stockholm, Sweden). Aug. 2022. <https://github.com/Qile0317/SoFoCompBio22>.
- D2. **Yang, Qile et al.** *Final Presentation: Hallucinating Scaffolds Team* iGEM at Berkeley All Hands 3 (Berkeley CA, USA). Dec. 2023.

Software

- S1. **Yang, Qile**. *Einops for R: Flexible, Powerful, and Readable Tensor Operations* The Comprehensive R Archive Network. 2025. <https://qile0317.github.io/einops/>.
- S2. **Yang, Qile**. *FastUtils: Fast, Readable Utility Functions* The Comprehensive R Archive Network. 2025. <https://qile0317.github.io/FastUtils/>.
- S3. Lin⁺, X., Lo⁺, K. & **Yang⁺, Qile**. *OKaiLoRa: a no-code platform for training, running, and sharing machine learning models in healthcare* Devpost. 2024. <https://devpost.com/software/okailora-ai>.

Selected Media Coverage

- 2025 Association of Single Cell Analysis (ASCA),  [scRepertoire 2: Enhanced and efficient toolkit for single-cell immune profiling \[J1\]](#)
- 2024 Chinese Software Developer Network, [Immune repertoire analysis - scRepertoire \[in Chinese\] \[J1\]](#)
- 2023 R-views (by RStudio), [May 2023: "Top 40" New CRAN Packages \[J2\]](#)

Selected Professional Organizations

2026–	External Relations Team Member, iGEM @ Berkeley
2025–	Operations & Tech Lead, Computational Biology @ Berkeley
2023–	Member, Bioconductor Developer Community
2025–	Member, Gerontological Society of America (GSA)
2025–	Member, The Antibody Society
2025–	Member, Boston Protein Design and Modelling Club (BPDMC)

Other Distinctions

2023	Swedish National Debate Team member, Swedish Schools Debating Organization
2023	World Schools Debating Championship Sponsorship (Withdrew due to external constraints), The Scholarship Fund in Memory of Ellinor Carlsson [In Swedish]

Relevant Coursework

Computational Biology: Probabilistic Modeling in Computational Biology | Computational Single-Cell and Systems Immunology | Machine Learning, Statistical Models, and Optimization for Molecular Problems | Computational Functional Genomics | Algorithms for Computational Biology

Machine Learning / Statistics / Maths: Introduction to Machine Learning | Introduction to Artificial Intelligence | Concepts of Probability | Linear Algebra and Differential Equations | Multi-variable Calculus | Discrete Mathematics and Probability Theory

Biology / Chemistry: General Genetics | Frontiers in Microbial Systems Biology | Chemical Structure and Reactivity | Organic Chemistry Laboratory | Introduction to Bioengineering | Ethics in Science and Engineering

Computer Science: Structure and Interpretation of Computer Programs | Data Structures | Introduction to Competitive Programming

Last updated: July 16, 2026